

# A Practitioner's Guide to Estimating School Effects in Centralized Assignment Systems<sup>\*</sup>

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*Centralized school assignment with lottery tie-breaking generates quasi-experiments that identify school effects, but existing methods require applicants' rank-order lists and lottery numbers, data that is not always available. I characterize which data from a match identifies a propensity score under which offers are ignorable. When each school draws its own lottery, offers at a school are randomly assigned among applicants rejected by every school they rank above it, so comparing applicants offered and rejected there requires only waitlists. Under a single lottery, this comparison is biased, and the order of each applicant's rejections is also needed. Rank-order lists identify the score without lottery numbers, while the assignment and published aggregates do not in general. I show how to compute each score in finite markets, how schools' waitlist orders reveal which of them share a lottery, and how to handle priorities and screened schools. In simulations, all valid designs recover the true effect, and richer data yields more precise estimates.*

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# 1 Introduction

A core problem in education research is estimating school effectiveness while accounting for the selection of students into schools. Many districts assign seats through deferred acceptance (DA) with lottery tie-breaking. [Abdulkadiroğlu et al. \(2017, 2022\)](#) show that match data from these districts identifies causal effects of attending a school: the match can be used to compute propensity scores, the probability that each applicant is offered a seat at each school. Following [Rosenbaum and Rubin \(1983\)](#), controlling for these scores compares applicants with the same chance of an offer, whose offers differ only because of the lottery.

Doing so requires rich data: rank-order lists, lottery numbers, capacities, and, at screened schools (which admit by test scores or other criteria), the scores used for screening. But this data is often inaccessible due to confidentiality requirements, record-keeping policies, or data degradation. For example, in other research with Chicago Public Schools, I have access to rank-order lists without lottery numbers. A district may instead release only the set of schools that rejected each applicant: the waitlists she is on.

Under what sets of information can a researcher recover propensity scores and then estimate school effects? I characterize which combinations of match data identify a propensity score for which offers are *ignorable* given the score — independent of potential outcomes given the score. The score need not condition on applicants' lists, only on information settled before an applicant reaches the school, which she does once every school she ranks above it has rejected her.

Using a continuum market model, I show that the information required to estimate causal effects depends on whether lottery tie-breakers are reused across schools. When each school draws its own lottery, as under multiple tie-breaking, offers among applicants who reach a school are randomly assigned, so causal identification requires only waitlists. When schools share a lottery, as under single tie-breaking, comparing applicants offered and rejected at a school is biased, and the order in which each applicant was rejected is also required. When a school uses its own lottery, lottery numbers alone yield an alternative design for estimating causal effects. Rank-order lists identify the score without lottery numbers whether or not schools share a lottery, while the assignment and published aggregates do not in general. Finally, waitlist order (each school's ordering of its applicants) alone reveals which schools share a lottery, so the researcher need not know this in advance. [Table 2](#) summarizes these results.

[Section 4](#) turns from identification to practice and computes each score in finite markets, and [Section 5](#) extends the methods to priority groups (such as preference for neighborhood applicants or siblings) and to screened schools. More data buys precision. In a market of 4,000 applicants with multiple tie-breaking, the root mean squared error of the estimated effect is 0.17 with lottery numbers alone, 0.09 with waitlists, and 0.06 with rank-order lists. Relative to estimates with rank-order lists, using lottery numbers alone requires about eight times as many applicants to achieve the same precision. [Section 7](#) turns these results into a step-by-step procedure.

Relative to [Abdulkadiroğlu et al. \(2017, 2022\)](#), who take lists and lottery numbers as given, I ask which subsets of that data suffice. Like [Borusyak and Hull \(2023\)](#), I exploit a known assignment process. The results also bear on the choice between single and multiple tie-breaking. While multiple tie-breaking can decrease student welfare ([Abdulkadiroğlu, Pathak and Roth, 2009](#); [Ashlagi](#)

and Nikzad, 2020), it also allows for simpler estimation of school effects.

## 2 Setup

Researchers and policymakers are often interested in the causal effect of attending a school  $s$  on an outcome, such as test scores, graduation, or behavioral measures. Formally, let  $D_i$  indicate that applicant  $i$  attends  $s$ , and let  $Y_{1i}$  and  $Y_{0i}$  be her potential outcomes if she does and does not attend, so that  $Y_i = Y_{0i}(1 - D_i) + Y_{1i}D_i$ . In a constant-effects model,

$$Y_i = \beta_s D_i + Y_{0i}, \quad (1)$$

where  $\beta_s = Y_{1i} - Y_{0i}$  is the effect of attending  $s$ . Regressing  $Y_i$  on  $D_i$  does not recover  $\beta_s$  due to omitted variable bias. Applicants who attend  $s$  differ from those who do not in ways that are correlated with their outcomes, such as through prior achievement or neighborhood location.

Data from centralized assignment systems can account for this selection bias. When a district assigns seats through lottery tie-breaking, applicants with the same preferences are offered seats at random. An offer of a seat at  $s$  is then a valid instrument for enrollment at  $s$ , after controlling for each applicant's probability of receiving it (Abdulkadiroğlu et al., 2017). Write  $D_{1i}$  and  $D_{0i}$  for applicant  $i$ 's potential enrollment: whether she would attend  $s$  if offered a seat there and if not.

### 2.1 School choice with lottery tie-breaking

Computing that probability requires a model of how the district assigns seats. Let  $I$  be a set of  $n$  applicants and let  $s \in \{0, 1, \dots, S\}$  index schools, with  $s = 0$  denoting the outside option. School  $s$  has  $q_s$  seats, and the outside option has  $q_0 > n$ , so that all applicants can choose the outside option. Each applicant submits a strict preference ordering over schools,  $\succ_i$ , where  $a \succ_i b$  means that  $i$  prefers school  $a$  to school  $b$ . These preferences define an applicant's *type*,  $\theta_i$ ; since there are finitely many schools, there are finitely many types.<sup>1</sup> Write  $B_{\theta s} = \{b \mid b \succ_{\theta} s\}$  for the set of schools that type  $\theta$  prefers to  $s$ .

Schools choose between applicants using *tie-breakers*.<sup>2</sup> There are  $V$  tie-breakers, indexed by  $v$ ; school  $s$  uses tie-breaker  $v(s)$ , and  $S_v$  is the set of schools that use tie-breaker  $v$ . Applicant  $i$ 's value of tie-breaker  $v$  is  $R_{iv}$ , and schools prefer applicants with smaller values. Write  $B'_{\theta s} = \{b \in S_v \mid b \succ_{\theta} s\}$  for the schools that type  $\theta$  prefers to  $s$  and that share tie-breaker  $v$ .

*Assumption 2.1.1 (Lottery tie-breaking).* Each  $R_{iv}$  is a lottery number, drawn from  $U[0, 1]$  independently across applicants and tie-breakers, and independently of types, potential enrollment, and potential outcomes.

Two designs are common. Under *single tie-breaking* (STB),  $V = 1$  and all schools use the same lottery number. Under *multiple tie-breaking* (MTB),  $V = S$  and  $v(s) = s$ , so each school draws a separate, independent lottery.

<sup>1</sup>In practice, schools may also have *priority groups* for certain sets of applicants, including siblings and those living nearby. Section 5.1 extends the results to priorities.

<sup>2</sup>Section 5.2 discusses non-lottery tie-breakers, such as test scores and interview evaluations.

Seats are assigned through student-proposing deferred acceptance (DA), introduced by [Gale and Shapley \(1962\)](#). In the first round of the algorithm, each applicant applies to her most-preferred school, and each school tentatively admits the applicants with the smallest values of its tie-breaker up to its capacity, and rejects the rest. In each later round, rejected applicants apply to their next-most-preferred school, and each school again tentatively admits up to capacity from the set of new applicants and those tentatively held. The algorithm ends when no applicant is rejected, resulting in a match  $\mu$ . Write  $\mu_i$  for applicant  $i$ 's assignment and  $Z_i(s) \equiv \mathbf{1}\{\mu_i = s\}$  for an indicator that  $i$  is offered a seat at school  $s$ . Applicant  $i$  *reaches*  $s$  if she applies to it in some round, that is, if she ranks  $s$  and every school she prefers to  $s$  rejects her.

## 2.2 Causal effects and identification

The causal effect of attendance on outcomes can be recovered using an instrument. Offers alone are not a valid instrument for enrollment: the probability of an offer depends on where an applicant applies, which may be correlated with her potential outcomes. Suppose, however, that the researcher can find a control set  $X_i$  such that potential enrollment and potential outcomes are independent of the offer given these controls. If so,  $Z_i(s)$  is a valid instrument for enrollment at  $s$  after controlling for  $X_i$ .<sup>3</sup> A two-stage least squares (2SLS) regression of  $Y_i$  on  $D_i$ , instrumenting with  $Z_i(s)$  and controlling for  $X_i$ , recovers  $\beta_s$ .

The control set  $X_i$  is often unwieldy, in which case propensity scores are a more practical control. In school choice, the natural control is each applicant's type,  $X_i = \theta_i$ , because given type, offers depend only on the lottery ([Abdulkadiroğlu et al., 2017](#), Proposition 1). However, nearly every applicant submits a different list, so controlling for type runs into the curse of dimensionality ([Hahn, 1998](#)). To address this, [Rosenbaum and Rubin \(1983\)](#) show that instead of using  $X_i$  directly, it suffices to control for the *propensity score*  $p_s(x) \equiv \mathbb{P}(Z_i(s) = 1 \mid X_i = x)$ , the probability that applicants with  $X_i = x$  receive an offer at  $s$ .

*Definition 2.2.1.* Offers at  $s$  are *ignorable*<sup>4</sup> given  $X_i$  if  $Z_i(s) \perp\!\!\!\perp (D_{0i}, D_{1i}, Y_{0i}, Y_{1i}) \mid X_i$ .

*Lemma 2.2.2* ([Rosenbaum and Rubin, 1983](#), Theorem 3). If offers at  $s$  are ignorable given  $X_i$ , they are ignorable given  $p_s(X_i)$ .

By [Lemma 2.2.2](#), it suffices to control for  $p_s(X_i)$  rather than  $X_i$ , bypassing the curse of dimensionality. The second stage is

$$Y_i = \beta_s D_i + \sum_p \alpha_2(p) \mathbf{1}\{p_s(X_i) = p\} + \eta_i, \quad (2)$$

and the first stage is

$$D_i = \gamma_s Z_i(s) + \sum_p \alpha_1(p) \mathbf{1}\{p_s(X_i) = p\} + v_i, \quad (3)$$

where both sums run over the values  $p$  of  $p_s(X_i)$  strictly between 0 and 1. As argued above, because offers are ignorable given  $p_s(X_i)$ , 2SLS estimation of [Equations \(2\) and \(3\)](#) recovers  $\beta_s$ . If there are

<sup>3</sup>Validity also requires relevance, that offers shift enrollment at  $s$ ; the exclusion restriction, that an offer affects outcomes only through enrollment at  $s$ ; and, with heterogeneous effects, monotonicity, that no applicant enrolls at  $s$  only when not offered a seat there. Random assignment delivers none of these by itself.

<sup>4</sup>This is the independence half of what [Rosenbaum and Rubin \(1983, §1.3\)](#) call *strongly ignorable* assignment. "Strongly ignorable" also requires overlap, where each applicant has a non-degenerate propensity score.

heterogeneous treatment effects, the estimate corresponds to a weighted average of effects for compliers with non-degenerate scores (Imbens and Angrist, 1994).

The same propensity score can be used to test for balance. Conditional on the score, characteristics fixed before assignment — such as baseline test scores or gender — should not differ between applicants who are and are not offered a seat at  $s$ .

### 2.3 Cutoffs and the large-market score

By Lemma 2.2.2, estimating the effect of  $s$  requires only each applicant's probability of an offer at  $s$ . Abdulkadiroğlu et al. (2017) show how to compute this score from the data used to run DA: preferences, lottery numbers, and capacities.

The match computed by DA can be summarized using only each school's cutoff, the lottery number of the marginal admitted applicant (Azevedo and Leshno, 2016). Let  $c_s$  be the largest tie-breaker value among applicants offered a seat at  $s$  if  $s$  is full, and  $c_s = 1$  otherwise. The DA assignment is *stable*: no applicant would rather attend a school that has a vacant seat or that seats someone with a larger value of its tie-breaker than hers. Hence applicant  $i$  is offered a seat at a school  $s$  she ranks if and only if

$$R_{iv(s)} \leq c_s \quad \text{and} \quad R_{iv(b)} > c_b \quad \text{for all } b \in B_{\theta s}. \quad (4)$$

Each applicant is therefore assigned to her favorite school among those whose cutoffs she clears. In a finite market, cutoffs depend on the realized lottery draw, so  $p_s(\theta)$  has no closed form.

Following Abdulkadiroğlu et al. (2017), I characterize the score in a *continuum economy*, a model of large markets: the set of applicants is  $I = [0, 1]$ , types are distributed according to  $F$  (statements about types refer to types in its support), and  $q_s$  is the proportion of  $I$  that can be seated at school  $s$ . DA is defined in the continuum through the same cutoffs, which are now constants rather than random.

If  $i$  is rejected from a set of schools during DA, those rejections are informative about her lottery numbers: a rejection at  $b$  means that her value of tie-breaker  $v(b)$  is larger than  $c_b$ . The *most informative disqualification* (MID) of tie-breaker  $v$  for type  $\theta$  at school  $s$  is the most lenient cutoff among the schools that  $\theta$  prefers to  $s$  and that use tie-breaker  $v$ :

$$MID_{\theta s}^v = \max \{c_b \mid b \in B_{\theta s}^v\}, \quad (5)$$

with  $MID_{\theta s}^v = 0$  when  $B_{\theta s}^v$  is empty (Abdulkadiroğlu et al., 2017). If she reaches  $s$ , her value of tie-breaker  $v$  must be larger than  $MID_{\theta s}^v$ .

*Theorem 2.3.1* (Abdulkadiroğlu et al., 2017, Theorem 1 and Appendix A.9; Abdulkadiroğlu et al., 2022, Appendix C.1). In a continuum economy satisfying Assumption 2.1.1, for all types  $\theta$  that rank  $s$ ,

$$p_s(\theta) = \prod_v (1 - MID_{\theta s}^v) \times \max \left( \frac{c_s - MID_{\theta s}^{v(s)}}{1 - MID_{\theta s}^{v(s)}}, 0 \right), \quad (6)$$

with  $p_s(\theta) = 0$  if  $MID_{\theta s}^v = 1$  for some  $v$ , and  $p_s(\theta) = 0$  for types that do not rank  $s$ .

The first factor in Equation (6) is the probability that a type- $\theta$  applicant is rejected by every school she prefers to  $s$ . The second is the probability that she is offered a seat at  $s$ , given those rejections. If a school she prefers to  $s$  has room for everyone ( $MID_{\theta s}^v = 1$ ), she never reaches  $s$ . Canceling  $1 - MID_{\theta s}^{v(s)}$  gives an equivalent form that is continuous in the cutoffs:

$$p_s(\theta) = \prod_{v \neq v(s)} (1 - MID_{\theta s}^v) \times \max(c_s - MID_{\theta s}^{v(s)}, 0). \quad (7)$$

By Assumption 2.1.1, the right side of Equation (7) is the probability of the event in Equation (4) at any cutoffs.

In the special cases of STB and MTB, the general formula for  $p_s(\theta)$  simplifies. Under STB, write  $MID_{\theta s} \equiv MID_{\theta s}^1 = \max\{c_b \mid b \in B_{\theta s}\}$ ; then Equation (6) reduces to

$$p_s(\theta) = (1 - MID_{\theta s}) \times \max\left(\frac{c_s - MID_{\theta s}}{1 - MID_{\theta s}}, 0\right) = \max(c_s - MID_{\theta s}, 0). \quad (8)$$

Under MTB,  $MID_{\theta s}^{v(s)} = 0$  and  $MID_{\theta s}^{v(b)} = c_b$  for  $b \in B_{\theta s}$ , so

$$p_s(\theta) = \prod_{b \in B_{\theta s}} (1 - c_b) \times c_s. \quad (9)$$

The identification results below rely on applicants who rank a school first, because their score there is the school's cutoff. I therefore assume that every school is someone's first choice.

*Assumption 2.3.2 (Rich support).* In the continuum economy, every school is ranked first by a positive proportion of applicants.

### 3 Data requirements to identify a propensity score

Theorem 2.3.1 shows how to compute propensity scores for  $s$  when cutoffs (derived from lottery numbers) and preferences are both known. This data is often unavailable; in this section, I show which data identifies a propensity score for  $s$ , and which does not. To do so, I use the continuum economy framework of Section 2.3. Suppose throughout this section that  $s$  is oversubscribed, so  $0 < c_s < 1$ .

A *data regime*  $\Omega$  is the collection of variables the researcher observes;  $\Omega_i$  denotes those pertaining to applicant  $i$  and  $P_\Omega$  their joint distribution. Unless otherwise stated, the researcher always observes capacities  $q$ , the assignment  $\mu$ , and the tie-breaker map  $v(\cdot)$ . The researcher may also observe the variables in Table 1.

The researcher may also observe characteristics  $G_i$  fixed before the match, such as baseline test scores, which are independent of lottery numbers. Whether  $i$  *qualifies* at school  $b$  is the indicator  $\mathbb{1}\{R_{iv(b)} \leq c_b\}$  that her lottery number clears its cutoff. Let  $\mathcal{I}_s \equiv (\theta_i, G_i, (\mathbb{1}\{R_{iv(b)} \leq c_b\})_{b \in B_{\theta_i}})$  collect  $i$ 's type, her characteristics, and whether she qualifies at each school she prefers to  $s$ .<sup>5</sup> These are the variables settled before  $i$  can apply to  $s$ ; they do not include her lottery number at  $s$ . Let  $A_i(s) \equiv \mathbb{1}\{\mu_i = s \text{ or } s \in W_i\}$  indicate that  $i$  reaches  $s$ ; by Equation (4),  $A_i(s)$  is a function of  $\mathcal{I}_s$ .

<sup>5</sup>For a type that does not rank  $s$ , let  $B_{\theta s}$  be the set of all schools it ranks.

Table 1: Match data beyond capacities, the assignment, and the tie-breaker map

| Symbol        | Name               | Description                                                                                                                      |
|---------------|--------------------|----------------------------------------------------------------------------------------------------------------------------------|
| $\mathcal{A}$ | aggregates         | published cutoffs and applicant counts                                                                                           |
| $r_i$         | assigned rank      | position of $\mu_i$ on $i$ 's list; missing if $i$ is unassigned                                                                 |
| $W_i$         | waitlists          | schools that rejected $i$ : those she ranked above $\mu_i$ (all ranked schools if she is unassigned), or the waitlists she is on |
| $W_i^o$       | ordered rejections | $W_i$ , ordered by $i$ 's preferences                                                                                            |
| $O_s$         | waitlist order     | school $s$ 's ordering of the applicants who applied to it                                                                       |
| $\theta_i$    | type               | $i$ 's rank-order list                                                                                                           |
| $R_i$         | lottery numbers    | $i$ 's value of each tie-breaker                                                                                                 |

Formally,  $W_i \equiv B_{\theta_i \mu_i}$ . Variables without a subscript  $i$ , such as  $W$  or  $R$ , denote the variable for all applicants.

*Definition 3.0.1.* Let  $\mathcal{M}$  be a class of continuum markets satisfying [Assumptions 2.1.1](#) and [2.3.2](#) in each of which  $s$  is oversubscribed,  $0 < c_s < 1$ . A data regime  $\Omega$  identifies a propensity score for  $s$  over  $\mathcal{M}$  if and only if there are functions  $g$ ,  $k$ , and  $h$  such that, in every market in  $\mathcal{M}$ :

- (i)  $K_i = k(X_i) \in \{0, 1\}$  selects a subsample of applicants;
- (ii) the controls are  $X_i = g(\Omega_i; P_\Omega)$ , where  $g(\Omega_i; P_\Omega)$  takes the same value for any two applicants with the same  $J_{is}$ ,<sup>6</sup> and, among applicants with  $K_i = 1$ , offers at  $s$  are ignorable given  $X_i$ ;
- (iii)  $p_s(X_i) = h(\Omega_i; P_\Omega)$ , so that  $h$  computes the propensity score from the data; and
- (iv) a positive proportion of applicants have  $K_i = 1$  and  $0 < p_s(X_i) < 1$ .

Applicants with  $K_i = 1$  and  $0 < p_s(X_i) < 1$  are offered a seat at  $s$  under some lottery draws and not others. They are *randomized at  $s$* , and form the estimation sample for [Equations \(2\) and \(3\)](#).

[Table 2](#) summarizes the results of this section, in the order of the subsections that follow.

Table 2: Data regimes and the propensity scores they identify for school  $s$ 

| Also observed       | $X_i$                                    | $p_s(X_i)$                    | Randomized at $s$                           | STB | MTB |
|---------------------|------------------------------------------|-------------------------------|---------------------------------------------|-----|-----|
| $\theta, R$         | $\theta_i$                               | <a href="#">Equation (6)</a>  | $0 < p_s(\theta_i) < 1$                     | Yes | Yes |
| $\theta$            | $\theta_i$                               | <a href="#">Equation (6)</a>  | $0 < p_s(\theta_i) < 1$                     | Yes | Yes |
| $W^o$               | $A_i(s), A_i(s) MID_{\theta_i s}^{v(s)}$ | <a href="#">Equation (11)</a> | $A_i(s) = 1, MID_{\theta_i s}^{v(s)} < c_s$ | Yes | Yes |
| $W$                 | $A_i(s)$                                 | $c_s A_i(s)$                  | $A_i(s) = 1$                                | No  | Yes |
| $\mathcal{A}, r, R$ | —                                        | —                             | —                                           | No  | No  |

Data is in addition to  $(q, \mu, v)$ . Entries report whether the regime identifies a propensity score for the offer at  $s$  ([Definition 3.0.1](#)), and the estimation sample; rows follow [Theorem 3.1.1](#) and [Propositions 3.1.3, 3.2.2, 3.3.1, 3.3.4, and 3.4.1](#). By [Corollary 3.5.2](#), the first three rows also hold with waitlist order  $O$  in place of  $v$ . Under STB, waitlists fail even with waitlist order and lottery numbers added ([Proposition 3.3.4](#)); if some schools share a lottery and others do not, the  $W$  row holds at  $s$  under the condition of [Proposition 3.3.1](#). With lottery numbers alone, qualification at a school with its own lottery gives an instrument with the compliers of the  $W$  row ([Proposition 3.4.2](#)).

### 3.1 Types

With types and lottery numbers, the researcher observes everything DA uses, which is the setting of [Abdulkadiroğlu et al. \(2017\)](#).

<sup>6</sup>This rules out conditioning on the match result  $\mu$  itself.

*Theorem 3.1.1* ([Abdulkadiroğlu et al., 2017](#), Theorem 1 and Appendix A.9; [Abdulkadiroğlu et al., 2022](#), Appendix C.1). The data  $(q, \mu, v, \theta, R)$  identifies a propensity score for  $s$  over any class of markets, with  $X_i = \theta_i$  and  $p_s(X_i)$  given by [Equation \(6\)](#). In that calculation, the cutoff  $c_b$  is the largest  $R_{iv(b)}$  among applicants offered a seat at  $b$  if  $b$  is full, and 1 otherwise.

Lottery numbers and the assignment reveal each cutoff, [Theorem 2.3.1](#) gives the score, and offers are ignorable given  $\theta_i$  by [Assumption 2.1.1](#). The rest of this section asks how much of this data can be dropped.

Some districts release rank-order lists but not lottery numbers, as in the Chicago example of [Section 1](#). Without lottery numbers, the cutoffs cannot be read off the assignment as the largest admitted lottery number, but they can be generated by simulating DA with new lottery draws.

An equivalent way to do so is to solve for cutoffs using lists and capacities.<sup>7</sup> Let  $p_s(\theta; c)$  denote the score in [Equation \(6\)](#) evaluated at cutoffs  $c$ . In the continuum, cutoffs describe a stable assignment if and only if they clear the market ([Azevedo and Leshno, 2016](#)):

$$\int p_s(\theta; c) dF(\theta) \leq q_s, \quad \text{with equality if } c_s < 1, \quad s = 1, \dots, S. \quad (10)$$

*Lemma 3.1.2.* In a continuum economy satisfying [Assumptions 2.1.1](#) and [2.3.2](#), [Equation \(10\)](#) has exactly one solution  $c \in [0, 1]^S$ .

*Proof.* See [Appendix A](#). □

Types and capacities therefore pin down the cutoffs, whichever schools share a lottery.<sup>8</sup>

*Proposition 3.1.3.* The data  $(q, \mu, v, \theta)$  identifies a propensity score for  $s$  over any class of markets, with  $X_i = \theta_i$  and  $p_s(X_i) = p_s(\theta_i; c)$ , the score of [Equation \(6\)](#) at the unique solution  $c$  of [Equation \(10\)](#). The score does not use the assignment  $\mu$ .

*Proof.* See [Appendix A](#). □

## 3.2 Ordered rejections

A district may release only the schools that rejected each applicant, in the order she ranked them, but neither preference lists nor lottery numbers. The standard lottery design compares applicants who win and lose a school's lottery ([Abdulkadiroğlu et al., 2011](#)). Within a DA match, the analog compares applicants offered a seat at  $s$  with those rejected there.

*Lemma 3.2.1.* In a continuum economy satisfying [Assumption 2.1.1](#), for every type  $\theta$  that ranks  $s$  and reaches it with positive probability,

$$\mathbb{P}(Z_i(s) = 1 \mid A_i(s) = 1, \theta_i = \theta) = \max \left( \frac{c_s - MID_{\theta_s}^{v(s)}}{1 - MID_{\theta_s}^{v(s)}}, 0 \right). \quad (11)$$

<sup>7</sup>When the assignment is observed, each cutoff is also the offer rate among applicants who rank the school first. Solving for cutoffs does not need the assignment.

<sup>8</sup>[Azevedo and Leshno \(2016, Theorem 1\)](#) obtain uniqueness either from full support of applicant types, which requires every combination of lottery numbers across schools and so fails whenever schools share a lottery, or from a regularity condition that gives uniqueness only for almost every vector of capacities. The proof of [Lemma 3.1.2](#) uses instead that lottery numbers are independent of types and across tie-breakers, together with rich support.



Further, the offer  $Z_i(s)$  is independent of  $(D_{0i}, D_{1i}, Y_{0i}, Y_{1i})$  conditional on  $A_i(s) = 1$  and  $\theta_i = \theta$ .

*Proof.* See [Appendix A](#). □

Rejections at more-preferred schools truncate the range of possible lottery numbers: reaching  $s$  truncates  $R_{iv(s)}$  from below at  $MID_{\theta, s}^{v(s)}$  and leaves it otherwise uniform. Under MTB, no school ranked above  $s$  shares its lottery, so there is no truncation.

*Proposition 3.2.2.* The data  $(q, \mu, v, W^v)$  identifies a propensity score for  $s$  over any class of markets, with  $X_i = (A_i(s), A_i(s)MID_{\theta, s}^{v(s)})$  and  $p_s(X_i)$  given by [Equation \(11\)](#) if  $A_i(s) = 1$  and 0 otherwise. Each cutoff is identified as the offer rate among applicants who rank the school first,  $c_b = \mathbb{P}(Z_i(b) = 1 \mid A_i(b) = 1, B_{\theta, b} = \emptyset)$ .

*Proof.* See [Appendix A](#). □

Ordered rejections reveal which rejections preceded  $s$ , and so which schools enter  $MID_{\theta, s}^{v(s)}$ .

### 3.3 Waitlists

Without the order of rejections, the researcher knows that  $i$  reached  $s$  but not which of her rejections came before it. The truncation in [Lemma 3.2.1](#) is then unknown, unless no school ranked above  $s$  shares its lottery.

*Proposition 3.3.1.* If no school other than  $s$  uses  $v(s)$ , then  $(q, \mu, v, W)$  identifies a propensity score for  $s$  over any class of markets in which this condition holds, with  $X_i = A_i(s)$  and  $p_s(X_i) = c_s A_i(s)$ .

*Proof.* See [Appendix A](#). □

Under this condition, offers among applicants who are offered a seat at  $s$  or rejected there are randomly assigned with probability  $c_s$ . The proof uses only a weaker condition, that no type that ranks  $s$  ranks a school with tie-breaker  $v(s)$  above  $s$ , but checking it requires applicants' lists.

*Corollary 3.3.2.* Over the class of MTB markets,  $(q, \mu, v, W)$  identifies a propensity score for every school, as in [Proposition 3.3.1](#).

*Proof.* See [Appendix A](#). □

Under STB, every school shares one lottery, so the condition fails whenever there is more than one school. The weaker condition then holds at  $s$  if and only if no type that ranks  $s$  ranks another school above it. When it fails, waitlists alone need not suffice. The following example, used for the rest of the section, shows why.

*Example 3.3.3.* Schools  $a$  and  $b$  have capacities 0.2 and 0.3. Types  $\theta_1$  ( $a \succ b$ ),  $\theta_2$  ( $b \succ a$ ), and  $\theta_3$  (only  $a$ ) each have mass  $1/3$ , and applicants have no other characteristics. Under STB,  $(c_a, c_b) = (0.3, 0.6)$ . By [Lemma 3.2.1](#), among applicants who reach  $b$ , a type- $\theta_1$  applicant is offered a seat there with probability  $(0.6 - 0.3)/(1 - 0.3) = 3/7$ ,

and a type- $\theta_2$  applicant with probability 0.6. Yet an applicant of either type who is rejected at both schools has  $W_i = \{a, b\}$ ; only the order of her rejections,  $(a, b)$  or  $(b, a)$ , tells the two apart. Under MTB,  $(c_a, c_b) \approx (0.24, 0.51)$ , and applicants of both types who reach  $b$  are offered a seat there with probability  $c_b$ .

Restricting to applicants whose waitlists reveal their score at  $b$  does not help, because which scores are revealed depends on the offer.<sup>9</sup>

Indeed, under STB no controls computed from waitlists make offers at  $b$  ignorable, even with more data added.

*Proposition 3.3.4.* The data  $(q, \mu, v, \mathcal{A}, r, R, W, O)$  does not identify a propensity score for  $s$  over the class of STB markets.

*Proof.* See [Appendix A](#). □

The proof uses the market of [Example 3.3.3](#) with  $s = b$ : a method that worked over the class of STB markets would have to work in this market, and none does. Unassigned applicants of types  $\theta_1$  and  $\theta_2$  with the same lottery number have the same waitlists,  $\{a, b\}$ , and the same position in each school's order, so the controls must treat them alike. The proof shows that this puts every applicant offered a seat at  $b$  in one cell, although among applicants who reach  $b$ , type  $\theta_1$  is offered a seat with probability  $3/7$  and type  $\theta_2$  with probability 0.6.<sup>10</sup> Adding aggregates, assigned ranks, and lottery numbers does not help, because the controls can use lottery numbers only through  $J_{ib}$ .

*Remark 3.3.5.* The designs randomize different applicants. The waitlist design randomizes only applicants offered a seat at  $s$  or rejected there. Controlling for types also randomizes applicants who rank  $s$  but may be seated at a school they prefer. With heterogeneous effects, the designs therefore estimate effects for different compliers, and richer data, by randomizing more applicants, gives more precise estimates ([Section 6](#)).

### 3.4 Assignments, aggregates, and lottery numbers

The sparsest data shows where applicants were placed, and possibly their lottery numbers, but not where they applied. It does not identify a propensity score, under any tie-breaking design.

*Proposition 3.4.1.* The data  $(q, \mu, v, \mathcal{A}, r, R)$  does not identify a propensity score for  $s$  over the class of MTB markets, or over the class of STB markets.

*Proof.* See [Appendix A](#). □

Under STB, this follows from [Proposition 3.3.4](#), because data that fails to identify a score still fails with variables removed. Under MTB, the proof again uses the market of [Example 3.3.3](#) with  $s = b$ . The data cannot separate applicants who lost the lottery at  $b$  from applicants who never applied

<sup>9</sup>An offered applicant reveals  $B_{\theta,b} = W_i$ , while a rejected one does so only if  $W_i = \{b\}$ , so the selection depends on the offer.

<sup>10</sup>If the researcher knew that no applicant ranks  $a$  above  $b$ , so that there were no type- $\theta_1$  applicants, waitlists would identify the score at  $b$ , because the weaker condition after [Proposition 3.3.1](#) would hold. Knowing this usually requires applicants' lists.

there. An unassigned type- $\theta_3$  applicant, who does not rank  $b$ , has the same data as unassigned applicants of types  $\theta_1$  and  $\theta_2$  with the same lottery numbers. Any control the data allows therefore puts unassigned type- $\theta_3$  applicants, who are never offered a seat at  $b$ , in the same cell as every applicant who is offered one. As under STB, lottery numbers do not help.

At a school with its own lottery, however, lottery numbers alone support a different design, which instruments with qualification at  $s$  itself,  $\mathbb{1}\{R_{iv(s)} \leq c_s\}$ , instead of the offer. Qualification does not depend on whether  $i$  ranks  $s$ , and it is computed from  $(q, \mu, v, R)$ , since  $c_s$  is the largest  $R_{iv(s)}$  among applicants offered a seat at  $s$ .

*Proposition 3.4.2.* If no school other than  $s$  uses  $v(s)$ , then qualification at  $s$  is independent of  $(\theta_i, A_i(s), D_{0i}, D_{1i}, Y_{0i}, Y_{1i})$ , it changes an applicant's assignment only by moving her to  $s$  from a school she ranks below  $s$ , and its compliers are applicants who reach  $s$  and enroll there if and only if offered a seat. If another school uses  $v(s)$ , offers at that school need not be independent of qualification at  $s$ .

*Proof.* See [Appendix A](#). □

At a school with its own lottery, qualification therefore has the same propensity score,  $c_s$ , for every applicant. It is then a valid instrument for enrollment at  $s$  on the full sample, with no controls, whenever the offer satisfies the exclusion restriction. [Table 2](#) still records lottery numbers alone as identifying no score, because  $c_s$  is the propensity score of qualification, not of the offer. Qualification's first stage is weaker than the offer's, because most applicants never reach  $s$ . When the lottery is shared, qualification satisfies the exclusion restriction only if the other schools on that lottery do not affect outcomes.

### 3.5 Tie-breakers

Suppose  $v$  is unknown. Schools  $s$  and  $s'$  *share applicants* if a positive proportion of applicants apply to both. Let  $\kappa_{ss'}$  be the probability that  $O_s$  and  $O_{s'}$  order the same way two randomly drawn common applicants, those who apply to both schools.

*Proposition 3.5.1.* For  $s$  and  $s'$  that share applicants,  $\kappa_{ss'} = 1$  if  $v(s) = v(s')$  and  $\kappa_{ss'} < 1$  otherwise. Since  $\kappa_{ss'}$  is a function of the distribution of waitlist order, the data  $(q, \mu, O)$  identifies whether two such schools share a tie-breaker.

*Proof.* See [Appendix A](#). □

The scores need only pairs of schools that share applicants, because a school  $b$  ranked above  $s$  enters  $p_s(\theta)$  only if applicants of type  $\theta$  can reach  $s$ , and those who do apply to both schools. Because only those pairs enter the scores, waitlist order can stand in for the tie-breaker map in each design that uses applicants' lists or the order of their rejections.

*Corollary 3.5.2.* Without the tie-breaker map, each of the following identifies a propensity score for  $s$  over any class of markets:

- (a)  $(q, \mu, O, \theta, R)$ , with the controls and score of [Theorem 3.1.1](#);
- (b)  $(q, \mu, O, \theta)$ , with the controls and score of [Proposition 3.1.3](#); and

(c)  $(q, \mu, O, W^o)$ , with the controls and score of [Proposition 3.2.2](#).

*Proof.* See [Appendix A](#). □

In finite markets, agreement on a few common applicants is weak evidence, which [Section 4.3](#) accounts for.

## 4 Implementation in finite markets

[Section 3](#) shows which data identifies a propensity score in the continuum, where cutoffs are constants. A district's match is finite: its cutoffs depend on the realized lottery draw, and the district may not report which schools share a lottery. This section turns the results of [Section 3](#) into estimators for such a match.

[Section 4.1](#) estimates each score from the data at hand, [Section 4.2](#) shows that, in large markets, the estimates approximate the propensity scores of the finite match, and [Section 4.3](#) recovers which schools share a lottery from waitlist order. The designs that use the offer's score all end the same way: 2SLS of [Equations \(2\) and \(3\)](#) on applicants whose estimated score is strictly between 0 and 1, instrumenting enrollment with the offer and controlling for a dummy for each value of the score.

### 4.1 Generating propensity scores

The scores of [Section 3](#) depend on the continuum cutoffs, which no finite match reveals. Each recipe below is the finite-market counterpart of the [Section 3](#) result named in its label, with the cutoffs replaced by estimates from the data at hand: realized cutoffs, solved cutoffs, or offer rates. Recipes (a)–(d) give the score of the offer and recipe (e) the score of qualification.

- (a) *Types and lottery numbers* ([Theorem 3.1.1](#)). Evaluate [Equation \(6\)](#) at the realized cutoffs  $\hat{c}_n$ , where  $\hat{c}_{ns}$  is the largest tie-breaker value among applicants offered a seat at  $s$  if  $s$  is full, and 1 otherwise.
- (b) *Types* ([Proposition 3.1.3](#)). Evaluate [Equation \(6\)](#) at the solved cutoffs  $\check{c}_n$ , the largest cutoffs that clear the market at the empirical distribution of types. With  $q_s$  counting seats, they solve

$$\sum_i p_s(\theta_i; c) \leq q_s, \quad \text{with equality if } c_s < 1, \quad s = 1, \dots, S. \quad (12)$$

By [Lemma 3.1.2](#), applied to the empirical distribution of types, they are the only solution whenever every school is some applicant's first choice. Simulating DA with new lottery draws also gives cutoffs, but each draw adds noise, which solving avoids.<sup>11</sup> The solved cutoffs can be computed one school at a time: given the other cutoffs, demand at  $s$ ,  $\sum_i p_s(\theta_i; c)$ , is continuous and piecewise linear in  $c_s$ , so each school's cutoff has a closed form, and a nonlinear Gauss–Seidel iteration computes  $\check{c}_n$ .

<sup>11</sup>In an economy where two applicants compete for one seat and four for two seats, both cells have a score of 1/2. Solved scores pool such cells exactly; scores simulated from repeated lottery draws do not.

*Lemma 4.1.1.* Start from  $c = (1, \dots, 1)$  and update the schools in turn, setting  $c_s$  to the largest value in  $[0, 1]$  at which  $\sum_i p_s(\theta_i; c) \leq q_s$ , given the other cutoffs. The cutoffs weakly decrease at every step and converge to  $\check{c}_n$ , the largest solution of Equation (12).

*Proof.* See Appendix A. □

- (c) *Ordered rejections* (Proposition 3.2.2). Evaluate Equation (11) at offer rates: estimate each  $c_b$ , including  $c_s$ , by the offer rate  $\bar{c}_{nb}$  among applicants who reach  $b$  and rank no school above it with tie-breaker  $v(b)$ , and compute  $\widehat{MID}_{\theta, s}^{v(s)}$  from  $W_i^o$  and these estimates. Restrict the sample to applicants with  $A_i(s) = 1$  and  $\widehat{MID}_{\theta, s}^{v(s)} < \bar{c}_{ns}$ . In small markets, these offer rates rest on few applicants, and the ratio in Equation (11) amplifies their errors.
- (d) *Waitlists* (Proposition 3.3.1). If no other school uses  $v(s)$ , the score is  $c_s$  for every applicant who reaches  $s$ , so estimate by 2SLS on applicants with  $A_i(s) = 1$ , with no further controls. Even in a finite market, whether  $i$  reaches  $s$  then does not depend on her lottery number at  $s$  (Lemma A.3).
- (e) *Lottery numbers* (Proposition 3.4.2). Qualification at the realized cutoff has the same score,  $\hat{c}_{ns}$ , for every applicant. At a school with its own lottery, estimate by 2SLS on the full sample, with no controls,

$$Y_i = \beta_s D_i + \alpha + \eta_i, \quad D_i = \gamma_s \mathbf{1}\{R_{iv(s)} \leq \hat{c}_{ns}\} + \alpha' + v_i, \quad (13)$$

instrumenting enrollment with qualification.<sup>12</sup>

## 4.2 Large markets

Each recipe uses a continuum score, at estimated cutoffs, in place of the propensity score of the finite match. The two agree in large markets. To make “large markets” precise, I follow Abdulkadiroğlu et al. (2017, 2022) and embed a finite market in an asymptotic sequence. The market of size  $n$  consists of the first  $n$  applicants of an i.i.d. sequence  $(\theta_i, G_i, R_i, D_{0i}, D_{1i}, Y_{0i}, Y_{1i})_{i \geq 1}$  drawn from the continuum economy, with capacities per applicant converging to  $q$ , and “almost surely” refers to this sequence. Let  $\mathbb{P}_n$  denote probabilities over lottery draws in the market of size  $n$ , given applicants’ types, and let  $p_{ns}(\theta) = \mathbb{P}_n(Z_i(s) = 1 \mid \theta_i = \theta)$  be the finite-market score of the offer.

Abdulkadiroğlu et al. (2017, Theorem 2) show that under single tie-breaking the score at the realized cutoffs approximates  $p_{ns}$ , and Abdulkadiroğlu et al. (2022, Theorem 2) show this in probability under general tie-breaking when all full schools have distinct cutoffs. The following proposition covers all five recipes of Section 4.1, with the score of the offer in parts (a)–(d) and the score of qualification in part (e).

*Proposition 4.2.1.* Suppose Assumptions 2.1.1 and 2.3.2 hold, and let  $c$  solve Equation (10). Along the asymptotic sequence, the following hold almost surely.

<sup>12</sup>Restricting to applicants known to have ranked  $s$ , such as those offered a seat there, would select on the lottery.

(a) *Types and lottery numbers.* The realized cutoffs converge,  $\hat{c}_n \rightarrow c$ , and

$$\sup_{\theta} |p_s(\theta; \hat{c}_n) - p_{ns}(\theta)| \rightarrow 0.$$

(b) *Types.* The solved cutoffs converge,  $\check{c}_n \rightarrow c$ , and

$$\sup_{\theta} |p_s(\theta; \check{c}_n) - p_{ns}(\theta)| \rightarrow 0.$$

(c) *Ordered rejections.* The offer rates converge,  $\bar{c}_{nb} \rightarrow c_b$  for every school  $b$ , and for every type  $\theta$  that reaches  $s$  with positive probability in the continuum,

$$\mathbb{P}_n(Z_i(s) = 1 \mid A_i(s) = 1, \theta_i = \theta) \rightarrow \max \left( \frac{c_s - MID_{\theta s}^{v(s)}}{1 - MID_{\theta s}^{v(s)}}, 0 \right).$$

(d) *Waitlists.* If no type that ranks  $s$  ranks a school with tie-breaker  $v(s)$  above  $s$ , then  $\mathbb{P}_n(Z_i(s) = 1 \mid A_i(s) = 1, \theta_i = \theta) \rightarrow c_s$  for every type  $\theta$  that reaches  $s$  with positive probability in the continuum.

(e) *Lottery numbers.* Qualification at the realized cutoff satisfies  $\mathbb{P}_n(R_{iv(s)} \leq \hat{c}_{ns} \mid \theta_i, D_{0i}, D_{1i}, Y_{0i}, Y_{1i}) \rightarrow c_s$ .

*Proof.* See [Appendix A](#). □

Each part follows from the convergence of the realized cutoffs to the continuum cutoffs, which carries over to the solved cutoffs and the offer rates; part (e) is the finite-market counterpart of [Proposition 3.4.2](#). Given types, simulated lottery numbers have the same distribution as the real ones, so part (a) also holds, in probability, for the score at simulated cutoffs.

As in [Abdulkadiroğlu et al. \(2017\)](#), standard errors treat the estimated score as known. With saturated controls, 2SLS depends on the score only through the partition of applicants into cells with a common score. Because the cutoffs converge, the estimated partition equals the limiting one in large markets, unless the limiting market has an exact tie, such as two equal cutoffs on one lottery or two equal scores. [Section 6](#) reports coverage in finite markets.

### 4.3 Recovering tie-breakers

When a district reports waitlist order but not the tie-breaker map, [Corollary 3.5.2](#) lets the order stand in for the map. In a finite market, compare the waitlist orders of each pair of schools  $s$  and  $s'$  on their common applicants, those who applied to both. If the two orders rank any two common applicants differently, the schools use different lotteries. If the orders agree on every common applicant, they may still agree by chance.

To bound that chance, group the common applicants by the schools that rejected them before they applied to the second of  $s$  and  $s'$ , which their lists or the order of their rejections reveal. Hold cutoffs fixed. If  $s$  and  $s'$  use different lotteries, the applicants in a group were rejected at the same cutoffs, so by [Assumption 2.1.1](#) their lottery numbers at each of the two schools are i.i.d., independently across the schools. The two orders then agree within every group with probability  $\pi_{ss'} = \prod_g 1/m_g!$ , where  $m_g$  is the size of group  $g$ . Join  $s$  and  $s'$  if  $w_{ss'} = -\log \pi_{ss'}$  is at least

$\bar{w} = \log(L/\varepsilon)$ , where  $L$  is the number of pairs compared, chosen without using the match (for example, all pairs). The probability that any pair with different lotteries is joined by chance is then at most  $\varepsilon$ . Finally, check with a topological sort that, within each connected component of joined schools, the union of the schools' orderings is acyclic; if it is not, remove joins in increasing order of weight until it is.

Without waitlist order, the match still places applicants offered a seat at each school above those rejected there, which can refute but not confirm a shared lottery.

## 5 Generalizations

The results so far assume that each school ranks all of its applicants by a lottery. In practice, schools may give priority to some applicants, such as siblings or those living nearby, and screened schools admit applicants by test scores or other criteria. Both extensions follow [Abdulkadiroğlu et al. \(2017, 2022\)](#).

### 5.1 Priorities

Let  $\rho_{\theta s} \in \{0, 1, \dots, J, \infty\}$  be type  $\theta$ 's priority at  $s$ , with smaller values better and  $\infty$  if  $\theta$  is ineligible; types now include priorities. For now, I assume each applicant is eligible for at most one priority group. School  $s$  admits applicants in increasing order of  $\rho_{\theta s} + R_{iv(s)}$ , so its cutoff  $c_s$  lies in  $[0, J + 1]$ . Its *marginal priority* is  $\rho_s = \lfloor c_s \rfloor$  and its *lottery cutoff* is  $\tau_s = c_s - \rho_s$ . An applicant who reaches  $s$  is always seated if  $\rho_{\theta s} < \rho_s$  ( $t_{\theta s} = a$ , always seated), seated if and only if  $R_{iv(s)} \leq \tau_s$  if  $\rho_{\theta s} = \rho_s$  ( $t_{\theta s} = c$ , conditionally seated), and never seated if  $\rho_{\theta s} > \rho_s$  ( $t_{\theta s} = n$ , never seated). Following [Abdulkadiroğlu et al. \(2017, 2022\)](#),

$$MID_{\theta s}^v = \begin{cases} 0 & \text{if } B_{\theta s}^v = \emptyset \text{ or } \rho_{\theta b} > \rho_b \text{ for all } b \in B_{\theta s}^v, \\ 1 & \text{if } \rho_{\theta b} < \rho_b \text{ for some } b \in B_{\theta s}^v, \\ \max\{\tau_b \mid b \in B_{\theta s}^v, \rho_{\theta b} = \rho_b\} & \text{otherwise,} \end{cases} \quad (14)$$

and

$$p_s(\theta) = \prod_v (1 - MID_{\theta s}^v) \times \begin{cases} 0 & \text{if } t_{\theta s} = n, \\ 1 & \text{if } t_{\theta s} = a, \\ \max\left(\frac{\tau_s - MID_{\theta s}^{v(s)}}{1 - MID_{\theta s}^{v(s)}}, 0\right) & \text{if } t_{\theta s} = c, \end{cases} \quad (15)$$

with  $p_s(\theta) = 0$  if  $MID_{\theta s}^v = 1$  for some  $v$ .

Transforming the data into a “pseudo-school” market accommodates applicants eligible for multiple priority groups. To do so, split each school into pseudo-schools  $s^0, \dots, s^J$ , one per priority group, all using  $v(s)$ , and let each applicant rank the pseudo-schools of her groups in place of  $s$ . In the continuum, pseudo-schools of groups better than  $\rho_s$  have cutoff 1, that of group  $\rho_s$  has cutoff  $\tau_s$ , and those of worse groups have cutoff 0, so [Equations \(14\) and \(15\)](#) are [Equations \(5\) and \(6\)](#) in the augmented economy.

The scores of [Section 3](#) then carry over, with three changes. Each data regime must record

applicants' priority groups: at  $s$  for the designs that use waitlists or ordered rejections, and also at the schools ranked above  $s$  for ordered rejections. Cutoffs are identified within priority groups:  $\tau_b$  is the offer rate in  $b$ 's marginal group, and [Assumption 2.3.2](#) must hold within each group, as in [Abdulkadiroğlu et al. \(2017, Assumption 1\)](#). And because [Lemma 3.1.2](#) covers only markets without priorities, the design with types but no lottery numbers reads its cutoffs from these offer rates instead of solving [Equation \(10\)](#).

## 5.2 Screened schools

Screened schools break ties with non-lottery tie-breakers, such as test scores, which may be correlated with potential outcomes. [Abdulkadiroğlu et al. \(2022\)](#) replace lottery randomization with local randomization near the cutoff. Given a bandwidth  $\delta$ , an applicant in the marginal priority group at a screened school is never seated there if her tie-breaker exceeds the cutoff by more than  $\delta$ , always seated if it falls more than  $\delta$  below, and conditionally seated otherwise. If the tie-breaker has a continuously differentiable distribution, conditionally seated applicants clear the cutoff with probability approaching  $1/2$  as  $\delta \rightarrow 0$ , independently of their potential outcomes. The classification parallels priorities: near its cutoff, each screened school behaves like a lottery school with cutoff  $1/2$ , and the resulting local DA propensity score replaces each relevant screened school's factor in [Equation \(6\)](#) with  $1/2$  ([Abdulkadiroğlu et al., 2022, Theorem 1](#)). That result needs distinct cutoffs at all full schools ([Abdulkadiroğlu et al., 2022, Assumption 2](#)). Estimation adds controls for the screened tie-breakers within a bandwidth  $\delta_n$  with  $\delta_n \rightarrow 0$  and  $n\delta_n \rightarrow \infty$  ([Abdulkadiroğlu et al., 2022, Theorem 2](#)).

The local score requires screened tie-breaker values, which no data regime in [Table 2](#) reveals. They are needed for offers at screened schools, and for the type-level score  $p_s(\theta)$  of applicants who rank a screened school above  $s$ . They are not needed for designs that condition on reaching  $s$ . Screened tie-breakers are independent of  $R_{iv(s)}$ , so [Equation \(11\)](#) holds as stated, and the rejection designs of [Propositions 3.2.2](#) and [3.3.1](#) and the qualification design at a school with its own lottery apply unchanged.

## 6 Simulations

I simulate two economies to show that the valid designs in [Table 2](#) recover school effects in finite markets, and to compare their precision. The first is Example 2 of [Abdulkadiroğlu, Angrist, Narita, and Pathak \(2017\)](#), hereafter AANP, replicated  $m$  times: schools  $a$ ,  $b$ , and  $c$  each have  $m$  seats, and  $m$  applicants of each of four types rank  $c$ ;  $c \succ b \succ a$ ;  $b \succ a$ ; and  $a$ . The target school is  $a$ ; under STB, its large-market scores there are  $0$ ,  $1/12$ ,  $1/12$ , and  $5/6$ . The second, the *random-preference economy*, has six schools and logit preferences, in which an unobserved ability raises outcomes and the utility of both the target school, which is of middling selectivity, and the most popular school. In both, the effect of enrolling at the target school is  $1$ , 85 percent of applicants offered a seat enroll, and no other school affects outcomes. Untreated outcomes differ by type in the first economy and with ability in the second, so comparisons that ignore selection are biased. For each economy, tie-breaking design, and data regime, I draw the market and its lotteries 1,000



times and estimate the effect by 2SLS as in [Section 4](#). [Appendix B](#) gives the exact specification.<sup>13</sup>

Table 3: Effect estimates by data regime

| Data                                                  | STB  |      |          | MTB  |      |          |
|-------------------------------------------------------|------|------|----------|------|------|----------|
|                                                       | Mean | SD   | Coverage | Mean | SD   | Coverage |
| <i>AANP (2017) Example 2, 250 applicants per type</i> |      |      |          |      |      |          |
| Offer, no controls                                    | 2.96 | 0.12 | 0.00     | 2.66 | 0.11 | 0.00     |
| Waitlists                                             | 1.88 | 0.17 | 0.00     | 1.00 | 0.16 | 0.94     |
| Ordered rejections                                    | 0.99 | 0.16 | 0.96     | 1.00 | 0.16 | 0.94     |
| Types and lottery numbers                             | 0.99 | 0.15 | 0.96     | 1.00 | 0.11 | 0.94     |
| Types, solved cutoffs                                 | 0.99 | 0.15 | 0.96     | 1.00 | 0.11 | 0.94     |
| Lottery numbers (qualification)                       | 0.98 | 0.49 | 0.96     | 1.00 | 0.32 | 0.95     |
| <i>Random preferences, 2,000 applicants</i>           |      |      |          |      |      |          |
| Offer, no controls                                    | 1.79 | 0.11 | 0.00     | 1.66 | 0.11 | 0.00     |
| Waitlists                                             | 1.18 | 0.14 | 0.70     | 1.00 | 0.13 | 0.94     |
| Ordered rejections                                    | 1.00 | 0.15 | 0.95     | 1.00 | 0.13 | 0.94     |
| Types and lottery numbers                             | 1.00 | 0.14 | 0.95     | 1.00 | 0.12 | 0.94     |
| Types, solved cutoffs                                 | 1.00 | 0.14 | 0.95     | 1.00 | 0.12 | 0.94     |
| Lottery numbers (qualification)                       | 1.01 | 0.40 | 0.94     | 1.00 | 0.23 | 0.95     |

Mean and standard deviation of 2SLS estimates of the effect of enrolling at the target school across 1,000 draws, and the share of draws whose 95 percent confidence interval, from robust standard errors, contains the true effect of 1. Example 2 of [Abdulkadiroğlu et al. \(2017\)](#) has 1,000 applicants and 250 seats per school; the random-preference economy has 2,000 applicants and 1,400 seats at six schools. “Offer, no controls” instruments enrollment with the offer on all applicants. The other rows follow [Section 4.1](#).

Ignoring selection is costly: in [Table 3](#), instrumenting with the offer without controls overstates the effect by 66 to 196 percent.

The order of rejections identifies effects under STB and MTB. Waitlists alone do so only under MTB; under STB they overstate the effect by 88 percent in Example 2 and 18 percent in the random-preference economy. With observable rank-order lists, 2SLS recovers the effect both with lottery numbers and with cutoffs solved from the lists alone, and the two give identical or nearly identical estimates. With saturated controls, only the partition of applicants into score cells matters, and it depends only on the order of the cutoffs. Qualification recovers the effect under MTB because the target school has its own lottery, and under STB only because no other school affects outcomes here ([Proposition 3.4.2](#)).

While all valid designs are centered on the true effect, richer data yields more precise estimates because it randomizes more applicants. Qualification uses every applicant, although most provide no variation; waitlists use the applicants who reach the target school; and lists add applicants who may be seated at schools they prefer ([Remark 3.3.5](#)).

[Figure 1](#) plots the root mean squared error of each valid design against market size. In Example 2 under MTB, with 4,000 applicants, it is 0.17 with lottery numbers, 0.09 with waitlists, and 0.06 with lists. Since the error falls with the square root of market size, matching the precision of lists with lottery numbers alone takes about eight times as many applicants. In the random-preference economy, lists add fewer applicants, and the gains over waitlists are smaller.

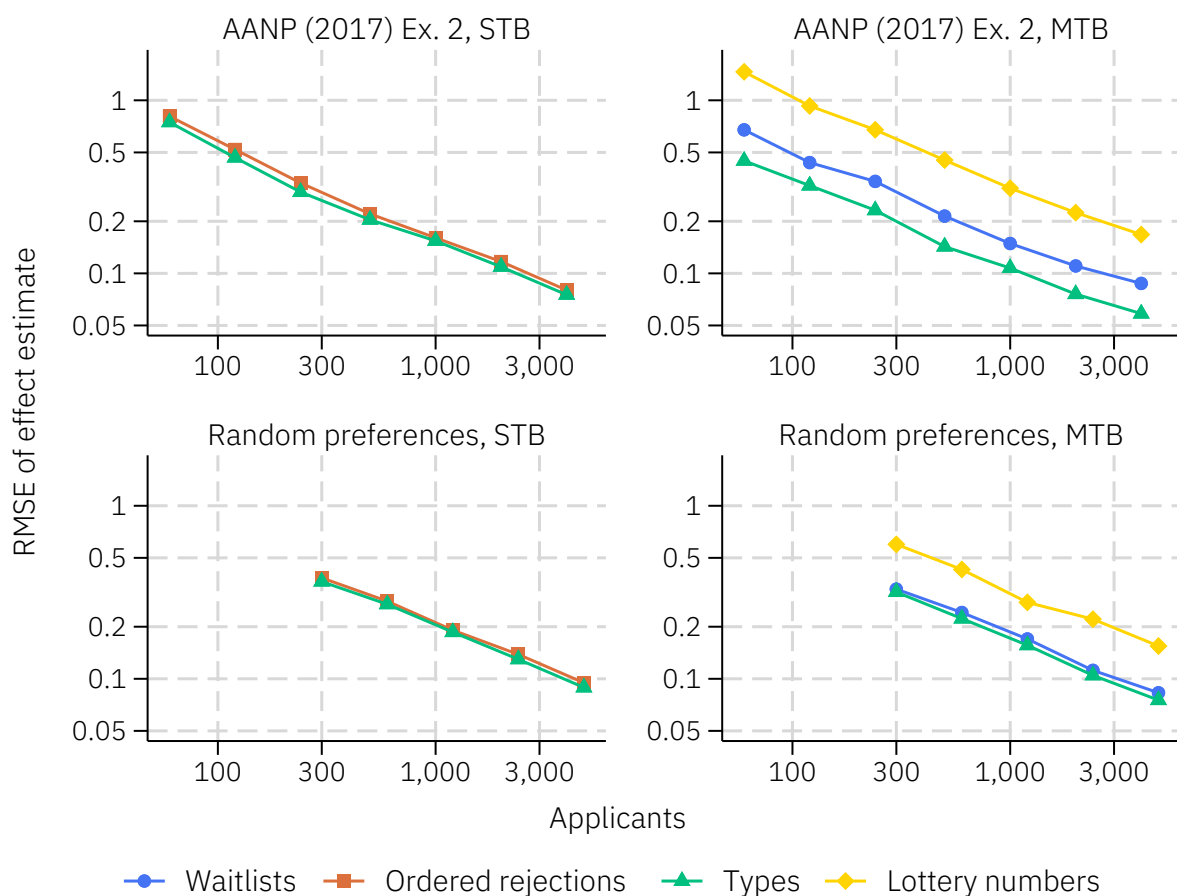
[Table B.1](#) in [Appendix B](#) repeats [Table 3](#) with 60 and 300 applicants. Estimates from the valid

<sup>13</sup>Code for all simulations is available from the author.

designs remain centered near 1. In the 60-applicant market, where score cells contain few applicants, coverage falls to 0.82–0.83 for ordered rejections and types under STB and to 0.90 for types under MTB, while waitlists and ordered rejections under MTB and qualification under either design stay at 0.93–0.96.

To test the procedure of [Section 4.3](#), I let schools 1–3 of the random-preference economy share one lottery, schools 4 and 5 share another, and school 6 have its own. With  $\varepsilon = 0.05$ , the procedure recovers this map in all 1,000 draws with 2,000 applicants and in 992 of 1,000 draws with 300, and it never joins schools that use different lotteries. Using the recovered map instead of the true one changes the mean and standard deviation of every estimate that uses the map by less than 0.001, and every such mean lies between 1.00 and 1.01.

Figure 1: Precision by data regime and market size



*Notes:* Root mean squared error of 2SLS estimates of the effect of enrolling at the target school across 300 draws per market size, for the designs that are valid under each tie-breaking design. Under STB, waitlists and lottery numbers alone do not identify a propensity score, so they are omitted. Under MTB, ordered rejections give the same estimates as waitlists, so only waitlists are shown. “Lottery numbers” is the qualification design. Both axes are logarithmic.

## 7 Practical recommendations

The results reduce to a procedure that estimates the effect of school  $s$  from whatever match data a district releases. Steps 1 and 2 prepare the data. Steps 3 and 4 choose a design that instruments with the offer, and step 5 estimates effects with it. Step 6 covers the one design that instruments with qualification instead.

1. *Remove priorities and classify applicants at screened schools.* If schools admit by priority group, split each school into one pseudo-school per group, all using the school's lottery (Section 5.1). At each screened school, classify each applicant as always, never, or conditionally seated from her screened tie-breaker and a bandwidth  $\delta$ ; conditionally seated applicants clear the cutoff with probability  $1/2$  (Section 5.2). The remaining steps then treat each pseudo-school as a school and each screened school as a lottery school with cutoff  $1/2$  for its conditionally seated applicants, and every regime must record applicants' priority groups. This classification requires screened tie-breakers; without them, no step identifies effects at a screened school.
2. *Find the tie-breakers.* If it is not known which schools share a lottery, recover it from waitlist order and applicants' lists or ordered rejections (Section 4.3). Without lists or ordered rejections, waitlist order still shows that two schools use different lotteries whenever it ranks two of their common applicants differently.
3. *Do you have rank-order lists?* If so, calculate propensity scores at  $s$  using Equation (6): at the realized cutoffs if lottery numbers are observed, and at the solved cutoffs otherwise (Section 4.1). With priorities and no lottery numbers, estimate the cutoffs by offer rates within priority groups instead of solving for them (Section 5.1). Then go to step 5. If not, go to step 4.
4. *Do you have waitlist data?* If so, restrict to applicants who reach  $s$ ,  $A_i(s) = 1$ .<sup>14</sup>
  - *Do you know the order of each applicant's rejections?* If so, the score is Equation (11), with cutoffs estimated by offer rates (Section 4.1). This works whether or not  $s$  shares its lottery.
  - *If not, does  $s$  have its own lottery?* If so, as under MTB, the score is the offer rate  $c_s$  (Section 4.1). If  $s$  shares its lottery, as under STB, waitlists do not identify a propensity score at  $s$  in general (Proposition 3.3.4).

If either case gives a score, go to step 5. Without waitlist data, go to step 6.

5. *Estimate effects.* Restrict the sample to applicants randomized at  $s$ . Estimate Equations (2) and (3) by 2SLS, instrumenting enrollment at  $s$  with the offer and controlling for a dummy for each value of the score and, with screened schools, for the screened tie-breakers within the bandwidth. Test balance by regressing baseline characteristics on the offer with the same controls.
6. *Do you have lottery numbers?* If so, and if  $s$  has its own lottery, estimate Equation (13) by 2SLS on the full sample of applicants (Section 4.1). In large markets every applicant qualifies with probability  $c_s$ , so no controls are needed. Otherwise, the data does not identify a propensity score for the offer at  $s$  in general (Proposition 3.4.1).

<sup>14</sup>At a lottery school, this step needs no screened tie-breakers, even when applicants rank screened schools above  $s$ : conditioning on reaching  $s$  absorbs them (Section 5.2).

Propensity scores from different steps randomize different applicants, so the resulting estimates average effects over different compliers ([Remark 3.3.5](#)). For districts, releasing the propensity score itself always enables estimation of school effects without releasing preferences or lottery numbers. Under MTB, waitlists alone suffice for causal identification, though more information increases precision.

Prior work has shown that MTB can decrease student welfare relative to STB ([Abdulkadiroğlu, Pathak and Roth, 2009](#); [Ashlagi and Nikzad, 2020](#)). However, MTB has a minor benefit for researchers: it identifies school effects from waitlists alone.

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## A Proofs

*Lemma A.1.* Fix any continuum economy, and write  $d_s(c) \equiv \int p_s(\theta; c) dF(\theta)$  for demand at  $s$ . Start from  $c = (1, \dots, 1)$  and update the schools in turn, setting  $c_s$  to the largest value in  $[0, 1]$  at which  $d_s(c) \leq q_s$ , given the other cutoffs. The cutoffs weakly decrease at every step and converge to a solution  $c^*$  of Equation (10) with  $c^* \geq c'$ , coordinate by coordinate, for every solution  $c'$ ; that is, to the largest solution.

*Proof.* I show that the cutoffs decrease and stay above every solution, and that their limit is a solution. By Equation (7),  $d_s$  is continuous, nondecreasing in  $c_s$ , nonincreasing in  $c_b$  for  $b \neq s$ , and zero at  $c_s = 0$ , so each update is well defined.

*The iterates decrease.* Number the updates, which cycle through schools  $1, \dots, S$ , and show by strong induction that each weakly lowers the cutoff it changes. Each of the first  $S$  updates starts from  $c_s = 1$ , so it weakly lowers the cutoff. At a later update of  $s$ , by the inductive hypothesis, each of the  $S - 1$  updates since the previous update of  $s$  weakly lowered the cutoff it changed, so demand at  $s$  is now weakly higher at every value of  $c_s$ . Every value above the old  $c_s$  put demand above capacity at the previous update, so it still does, and the new  $c_s$  is weakly below the old one.

*The iterates stay above every solution.* Let  $c'$  be a solution, and suppose that  $c' \leq c$  before an update of  $s$ , as holds at the start. Because the other cutoffs are weakly above  $c'_{-s}$ , demand at  $s$  at  $c_s = c'_s$  is at most  $d_s(c') \leq q_s$ , so the update sets  $c_s \geq c'_s$ , and  $c' \leq c$  still holds.

*The limit is the largest solution.* The cutoffs decrease to a limit  $c^*$ , and  $c^* \geq c'$  for every solution  $c'$ . Demand at  $s$  is at most  $q_s$  just after each update of  $s$ , so  $d_s(c^*) \leq q_s$  by continuity. If  $c_s^* < 1$ , then for each  $x \in (c_s^*, 1]$ , late updates of  $s$  set  $c_s < x$ , so  $d_s(x, c_{-s}) > q_s$  there, and  $d_s(x, c_{-s}^*) > q_s$  because  $c_{-s} \geq c_{-s}^*$ ; letting  $x \downarrow c_s^*$  gives  $d_s(c^*) = q_s$ . So  $c^*$  solves Equation (10) and is the largest solution.  $\square$

*Proof of Lemma 3.1.2.* A solution exists by Lemma A.1. Let  $c$  and  $c'$  be solutions, and suppose, for contradiction, that the set  $T = \{t : c_t < c'_t\}$  of schools that are more lenient at  $c'$  is nonempty. I show that the schools in  $T$  would hold more applicants at  $c'$  than at  $c$ , which is impossible because they are already full at  $c$ . By Assumption 2.1.1, Equation (7) is the probability of the event in Equation (4) at any cutoffs, so  $d_s$  evaluated at any cutoffs is the mass of applicants assigned to  $s$  when each applicant is assigned to her favorite school among those whose cutoffs she clears. Let  $M$  and  $M'$  be the sets of applicants assigned to a school in  $T$  at  $c$  and at  $c'$ .

First,  $M \subseteq M'$ . An applicant assigned to  $t \in T$  at  $c$  still clears  $t$  at  $c'$ , where  $t$  is more lenient. Every school she prefers to  $t$  rejects her at  $c$ , and those outside  $T$ , which are weakly stricter at  $c'$ , reject her at  $c'$  as well. So at  $c'$  she is assigned to  $t$  or to a school in  $T$  that she prefers to  $t$ .

Second,  $M' \setminus M$  has positive mass. Let  $t^*$  be a school in  $T$  with the largest cutoff at  $c$ . Applicants who rank  $t^*$  first, have  $R_{iv(t^*)} \in (c_{t^*}, c'_{t^*}]$ , and have  $R_{iv} > c_{t^*}$  for every other tie-breaker  $v$  have positive mass, by Assumptions 2.1.1 and 2.3.2 and because  $c_{t^*} < c'_{t^*} \leq 1$ . At  $c'$ , they are assigned to their first choice,  $t^*$ , and at  $c$ , they fail every school in  $T$ , because  $c_t \leq c_{t^*}$  for each  $t \in T$ . So they lie in  $M' \setminus M$ .

By Equation (10), each  $t \in T$  is full at  $c$ , because  $c_t < c'_t \leq 1$ , so  $M$  has mass  $\sum_{t \in T} q_t$ . Since  $M \subseteq M'$  and  $M' \setminus M$  has positive mass,  $M'$  has mass greater than  $\sum_{t \in T} q_t$ , so some school in  $T$  would exceed its capacity at  $c'$ , contradicting Equation (10). Hence  $T$  is empty, so  $c \geq c'$ , and, by symmetry,  $c = c'$ .  $\square$

*Proof of Proposition 3.1.3.* DA is stable, so its cutoffs solve Equation (10) (Azevedo and Leshno, 2016, Lemma 1). By Lemma 3.1.2, they are its only solution, so the researcher can compute them from  $q$ ,  $v$ , and the distribution of types, without  $\mu$ . By Theorem 2.3.1,  $p_s(\theta_i; c)$  is the offer probability of type  $\theta_i$ , and offers are ignorable given  $\theta_i$  by Assumption 2.1.1. Take  $K_i = 1$  for all  $i$ . Applicants who rank  $s$  first, a positive mass by Assumption 2.3.2, have score  $c_s \in (0, 1)$ . So  $K_i = 1$  and  $X_i = \theta_i$ , which is part of  $\mathcal{I}_{is}$ , satisfy Definition 3.0.1 with score  $p_s(\theta_i; c)$  in every market, and none of these uses  $\mu$ .  $\square$

*Proof of Lemma 3.2.1.* Write  $U_i = (D_{0i}, D_{1i}, Y_{0i}, Y_{1i})$ . By Equation (4), each applicant is assigned to her favorite school among those whose cutoffs she clears, so an appli-

cant of type  $\theta$  reaches  $s$  if and only if she clears no school in  $B_{\theta s}$ , that is,  $R_{iv} > MID_{\theta s}^v$  for every tie-breaker  $v$ ; she is then offered a seat at  $s$  if and only if  $R_{iv(s)} \leq c_s$ . By [Assumption 2.1.1](#), given  $\theta_i = \theta$  and  $U_i$ , the lottery numbers are independent  $U[0, 1]$  draws, so given also  $A_i(s) = 1$ ,  $R_{iv(s)}$  is uniform on  $(MID_{\theta s}^{v(s)}, 1]$ , an interval of positive length because  $\theta$  reaches  $s$  with positive probability. So  $\mathbb{P}(Z_i(s) = 1 \mid A_i(s) = 1, \theta_i = \theta, U_i)$  is the right side of [Equation \(11\)](#), which does not depend on  $U_i$ . This gives [Equation \(11\)](#) and, because  $Z_i(s)$  is binary, the independence of  $Z_i(s)$  and  $U_i$  given  $A_i(s) = 1$  and  $\theta_i = \theta$ .  $\square$

*Proof of Proposition 3.2.2.* Take  $K_i = A_i(s)$  and  $X_i = (A_i(s), A_i(s)MID_{\theta_i s}^{v(s)})$ , which depends only on  $\mathcal{I}_{is}$ , because  $A_i(s)$  does, by [Equation \(4\)](#), and  $MID_{\theta_i s}^{v(s)}$  depends only on  $\theta_i$  and the cutoffs. By [Lemma 3.2.1](#), among applicants who reach  $s$ , the offer depends on the type only through  $MID_{\theta_i s}^{v(s)}$  and does not depend on  $(D_{0i}, D_{1i}, Y_{0i}, Y_{1i})$ , so offers are ignorable given  $X_i$  among applicants with  $K_i = 1$ , and the score is given by [Equation \(11\)](#) if  $A_i(s) = 1$  and is 0 otherwise, because an offer at  $s$  requires reaching  $s$ . Each cutoff  $c_b$  is identified as the offer rate at  $b$  among applicants who rank  $b$  first, by [Lemma 3.2.1](#); these applicants have positive mass by [Assumption 2.3.2](#), and the data reveals them as the applicants offered a seat at  $b$  with  $W_i$  empty and those for whom  $b$  comes first in  $W_i^o$ . For an applicant who reaches  $s$ , the data also reveals  $B_{\theta_i s}$ , which is  $W_i$  if  $\mu_i = s$  and the schools listed before  $s$  in  $W_i^o$  otherwise, and so  $X_i$  and its score. Finally, applicants who rank  $s$  first, a positive mass by [Assumption 2.3.2](#), reach  $s$  and have  $MID_{\theta_i s}^{v(s)} = 0$ , so their score is  $c_s \in (0, 1)$ . Hence  $K_i, X_i$ , and this score satisfy [Definition 3.0.1](#) in every market.  $\square$

*Proof of Proposition 3.3.1.* Take  $X_i = K_i = A_i(s)$ , which the data reveals and which depends only on  $\mathcal{I}_{is}$ , by [Equation \(4\)](#). Because no other school uses  $v(s)$ , rejections above  $s$  do not truncate  $R_{iv(s)}$ , that is,  $MID_{\theta s}^{v(s)} = 0$  for every type  $\theta$  that ranks  $s$ . So, by [Lemma 3.2.1](#), every applicant who reaches  $s$  is offered a seat there with probability  $c_s$ , whatever her type and potential outcomes. Offers are therefore ignorable given  $X_i$  among applicants with  $K_i = 1$ , with  $p_s(X_i) = c_s A_i(s)$  because an offer at  $s$  requires reaching  $s$ , and the cutoff is identified as the offer rate among applicants who reach  $s$ ,  $c_s = \mathbb{P}(Z_i(s) = 1 \mid A_i(s) = 1)$ . By [Assumption 2.3.2](#), a positive mass ranks  $s$  first and so reaches  $s$ , and their score  $c_s$  lies in  $(0, 1)$  because  $s$  is oversubscribed. Hence  $K_i = X_i = A_i(s)$  and  $p_s(X_i) = c_s A_i(s)$  satisfy [Definition 3.0.1](#) in every market in which no other school uses  $v(s)$ .  $\square$

*Proof of Corollary 3.3.2.* Under MTB, no two schools share a tie-breaker, so the condition of [Proposition 3.3.1](#) holds at every school.  $\square$

[Propositions 3.3.4](#) and [3.4.1](#) are negative results. Both use the market of [Example 3.3.3](#) with  $s = b$ , which satisfies [Assumptions 2.1.1](#) and [2.3.2](#) under both designs and in which  $b$  is oversubscribed. A data regime that identifies a propensity score over the class of STB or MTB markets must therefore satisfy [Definition 3.0.1](#) in this market under that design, and both proofs show that it cannot, using the following lemma.

*Lemma A.2.* Take the market of [Example 3.3.3](#) under STB or MTB, and let both potential outcomes  $Y_{0i}$  and  $Y_{1i}$  equal 0 for type  $\theta_1$ , 1 for type  $\theta_2$ , and 2 for type  $\theta_3$ ; call this value the applicant's outcome. Suppose that a data regime gives unassigned applicants of types  $\theta_1$  and  $\theta_2$  with the same lottery numbers the same data, and that  $g$ ,  $k$ , and  $h$  satisfy [Definition 3.0.1](#) for it in this market with  $s = b$ . Then all type- $\theta_2$  applicants and all type- $\theta_1$  applicants who fail to clear  $a$  have the same value  $x$  of  $X_i$ , and among applicants with  $X_i = x$ , the share offered a seat at  $b$  is the same for every outcome that has positive mass among them.

*Proof.* Both types are unassigned when they fail to clear both schools, which happens with positive probability under either design. By condition (ii), applicants with the same data, or with the same  $\mathcal{J}_{ib}$ , have the same  $X_i$ . Since unassigned applicants of the two types with the same lottery numbers have the same data, and all type- $\theta_2$  applicants share one value of  $\mathcal{J}_{ib}$ , as do all type- $\theta_1$  applicants who fail to clear  $a$ , all of these applicants have the same value  $x$  of  $X_i$ . They include every applicant offered a seat at  $b$ , so every other value of  $X_i$  has score 0, and condition (iv) requires  $K_i = 1$  when  $X_i = x$ . Condition (ii) then requires offers at  $b$  to be independent of potential outcomes among applicants with  $X_i = x$ , so, since the outcome is a function of potential outcomes, the share of these applicants offered a seat at  $b$  is the same for every outcome.  $\square$

*Proof of Proposition 3.3.4.* Under STB, an unassigned applicant of type  $\theta_1$  or  $\theta_2$  has  $W_i = \{a, b\}$ ,  $r_i$  missing, and a place in each school's order that depends only on  $R_i$ , so unassigned applicants of the two types with the same  $R_i$  have the same data. By [Lemma A.2](#), it suffices to show that, among applicants with  $X_i = x$ , two outcomes are offered a seat at  $b$  at different rates. The outcome-1 applicants, who are the type- $\theta_2$  applicants, are offered a seat at  $b$  at rate  $c_b = 0.6$ . The outcome-0 applicants are of type  $\theta_1$ , and only those who fail to clear  $a$  can be offered a seat at  $b$ , so among outcome-0 applicants with  $X_i = x$ , the share offered one is at most  $\mathbb{P}(R_i \leq 0.6 \mid R_i > 0.3) = 3/7 < 0.6$ .  $\square$

*Proof of Proposition 3.4.1.* Under STB, functions  $g$ ,  $k$ , and  $h$  that satisfied [Definition 3.0.1](#) for  $(q, \mu, v, \mathcal{A}, r, R)$  would satisfy it for the larger data of [Proposition 3.3.4](#), which is impossible. Under MTB, the data is  $(\mu_i, r_i, R_i)$  up to constants, and applicants with  $R_{ia} > c_a$  and  $R_{ib} > c_b$  are unassigned whatever their type, so applicants of all three types with the same such lottery numbers have the same data. By [Lemma A.2](#), it suffices to show that, among applicants with  $X_i = x$ , two outcomes are offered a seat at  $b$  at different rates. Here type- $\theta_1$  applicants who fail to clear  $a$  are offered a seat at  $b$  at the same rate as type- $\theta_2$  applicants,  $c_b$ , so these two types need not give different rates; I use type  $\theta_3$ , which does not rank  $b$ . The type- $\theta_3$  applicants who fail to clear  $a$  share one value of  $\mathcal{J}_{ib}$ , and some of them have the same data as unassigned type- $\theta_2$  applicants, so they too have  $X_i = x$ . They have outcome 2 and positive mass, and no type- $\theta_3$  applicant is offered a seat at  $b$ , so outcome-2



applicants with  $X_i = x$  are offered one at rate 0, while outcome-1 applicants are offered one at rate  $c_b > 0$ .  $\square$

This completes the negative results. The remaining proofs for [Section 3](#) concern the qualification design and unknown tie-breakers.

*Proof of [Proposition 3.4.2](#).* By [Equation \(4\)](#),  $Z_i(s) = A_i(s)\mathbb{1}\{R_{iv(s)} \leq c_s\}$ . If only  $s$  uses  $v(s)$ , then  $A_i(s)$ , and  $i$ 's assignment when  $Z_i(s) = 0$ , depend only on her type and the other tie-breakers, while qualification at  $s$  is a function of  $R_{iv(s)}$ . By [Assumption 2.1.1](#), qualification is therefore independent of  $(\theta_i, A_i(s), D_{0i}, D_{1i}, Y_{0i}, Y_{1i})$ . If  $A_i(s) = 0$ , qualification leaves  $i$ 's assignment unchanged; if  $A_i(s) = 1$ , it moves her to  $s$  from the school she is assigned to when rejected there, which she ranks below  $s$ . So its compliers are applicants who reach  $s$  and enroll there if and only if offered a seat. For the second claim, take STB in [Example 3.3.3](#) with  $s = b$ . A type- $\theta_1$  applicant qualifies at  $b$  if and only if  $R_i \leq 0.6$  and is offered a seat at  $a$  if and only if  $R_i \leq 0.3$ , so her probability of an offer at  $a$  is  $1/2$  if she qualifies at  $b$  and 0 if she does not. Hence offers at  $a$ , which uses the same lottery as  $b$ , are not independent of qualification at  $b$ .  $\square$

*Proof of [Proposition 3.5.1](#).* If  $v(s) = v(s')$ , both schools order common applicants by the same lottery number, so  $\kappa_{ss'} = 1$ . Otherwise, two randomly drawn common applicants have the same type with positive probability, because there are finitely many types. For two applicants of the same type  $\theta$ , applying to both schools restricts each lottery number separately ( $R_{iv(b)} > c_b$  for every school  $b$  that  $\theta$  prefers to  $s$  or to  $s'$ ), so, given that both apply, their values of any one tie-breaker are i.i.d., and different tie-breakers are independent. The two schools, which use different tie-breakers, then order them by independent fair coin flips, which disagree with probability  $1/2$ , so  $\kappa_{ss'} < 1$ .  $\square$

*Proof of [Corollary 3.5.2](#).* As in [Proposition 3.2.2](#), each cutoff is the offer rate among applicants who rank the school first, and each of the three data sets reveals these applicants, so the cutoffs are identified without  $v$ . Given the cutoffs, the controls and score of an applicant whose type  $\theta$  reaches  $s$  with positive probability depend on  $v$  only through which pairs of schools in  $B_{\theta s} \cup \{s\}$  share a tie-breaker. Type- $\theta$  applicants who reach  $s$ , a positive mass, apply to every school in this set, so each such pair shares applicants, and [Proposition 3.5.1](#) identifies whether it shares a tie-breaker. An applicant whose type reaches  $s$  with probability 0 never reaches  $s$ , and her controls and score do not depend on  $v$ . So each data set gives the controls and score of [Theorem 3.1.1](#), [Proposition 3.1.3](#), or [Proposition 3.2.2](#).  $\square$

*Proof of [Lemma 4.1.1](#).* Dividing demand and capacity by  $n$  does not change the update, and it turns [Equation \(12\)](#) into [Equation \(10\)](#) for the continuum economy whose type distribution is the empirical distribution of types, with capacities  $q_s/n$ . The iteration is therefore the one in [Lemma A.1](#) for that economy, and [Lemma A.1](#) gives the result.  $\square$

Section 4.1 uses the following lemma for the waitlist design.

*Lemma A.3.* In any finite market, if no school that applicant  $i$  ranks above  $s$  uses  $v(s)$ , then, holding fixed all types and every tie-breaker value other than  $R_{iv(s)}$ ,  $A_i(s)$  does not depend on  $R_{iv(s)}$ .

*Proof.* Until  $i$  proposes to  $s$ , DA does not use  $R_{iv(s)}$ , because it compares  $i$  with other applicants only at schools she has proposed to, and by assumption none of them uses  $v(s)$ . So DA runs identically under any value of  $R_{iv(s)}$  until  $i$  proposes to  $s$ , and whether she ever proposes to  $s$ , which is  $A_i(s)$ , does not depend on  $R_{iv(s)}$ .  $\square$

The proof of Proposition 4.2.1 rests on one lemma. Along the asymptotic sequence of Section 4.2, let  $\hat{c}_n$  be the realized cutoffs.

*Lemma A.4.* Suppose Assumptions 2.1.1 and 2.3.2 hold, and let  $c$  be the solution of Equation (10), which is unique by Lemma 3.1.2. Then  $\hat{c}_n \rightarrow c$  almost surely; for every school  $s$  and every type  $\theta$  that ranks  $s$ ,

$$\mathbb{P}_n(A_i(s) = 1 \mid \theta_i = \theta) \rightarrow \prod_v (1 - MID_{\theta s}^v);$$

and for every type  $\theta$ ,  $p_{ns}(\theta) \rightarrow p_s(\theta; c)$ , all almost surely.

*Proof.* The realized cutoffs converge, and given the cutoffs, an applicant's offer depends only on her own type and lottery numbers. By Lemma 3.1.2 and Azevedo and Leshno (2016, Lemma 1), the continuum economy has a unique stable matching, with cutoffs  $c$ . By Varadarajan's theorem, the empirical distribution of applicants' types and lottery numbers converges weakly to that of the continuum economy almost surely, and capacity per applicant converges to  $q$ , so the markets converge to the continuum economy in the sense of Azevedo and Leshno (2016); their strict-preference assumption holds because lottery numbers have no atoms. By Azevedo and Leshno (2016, Theorem 2),  $\hat{c}_n \rightarrow c$  almost surely. Fix a type  $\theta$  that ranks  $s$  and an applicant  $i$  of that type; for other types,  $A_i(s) = Z_i(s) = 0$  and  $p_s(\theta; c) = 0$ . By Equation (4) in the market of size  $n$ ,  $i$  reaches  $s$  if and only if  $R_{iv(b)} > \hat{c}_{nb}$  for all  $b \in B_{\theta s}$ , and is offered a seat there if, in addition,  $R_{iv(s)} \leq \hat{c}_{ns}$ . Since  $\hat{c}_n \rightarrow c$  and each  $R_{iv}$  differs from each  $c_b$  almost surely, these indicators converge almost surely to the same indicators at  $c$ . Conditioning on the first  $n$  types, as  $\mathbb{P}_n$  does, is the same as conditioning on all types, because the indicators depend only on the first  $n$  applicants, who are independent of the rest. So by the dominated convergence theorem for conditional expectations, the conditional probabilities converge almost surely to the probabilities of the events at  $c$ , which are  $\prod_v (1 - MID_{\theta s}^v)$  and, as noted after Equation (7),  $p_s(\theta; c)$ .  $\square$

*Proof of Proposition 4.2.1.* (a) By Lemma A.4,  $\hat{c}_n \rightarrow c$  and  $p_{ns}(\theta) \rightarrow p_s(\theta; c)$  almost surely. Since Equation (7) is continuous in the cutoffs,  $p_s(\theta; \hat{c}_n) \rightarrow p_s(\theta; c)$  almost surely as well. By the triangle inequality,  $p_s(\theta; \hat{c}_n) - p_{ns}(\theta) \rightarrow 0$  almost surely, uniformly in  $\theta$  because there are finitely many types.

(b) By the strong law of large numbers, the empirical distribution  $F_n$  of types satisfies  $F_n(\theta) \rightarrow F(\theta)$  almost surely for each of the finitely many types, so demand under  $F_n$ ,  $\sum_{\theta} F_n(\theta) p_s(\theta; c)$ , converges to demand under  $F$ ,  $d_s(c) = \sum_{\theta} F(\theta) p_s(\theta; c)$ , uniformly in  $c$ , because  $0 \leq p_s \leq 1$ . Dividing Equation (12) by  $n$ , demand under  $F_n$  at  $\check{c}_n$  is at most capacity per applicant, which converges to  $q_s$ , with equality if  $\check{c}_{ns} < 1$ . Let  $\tilde{c}$  be a limit point of  $\check{c}_n$ . Along a subsequence converging to  $\tilde{c}$ , demand at  $\check{c}_n$  converges to  $d_s(\tilde{c})$ , because  $d_s$  is continuous by Equation (7). So  $d_s(\tilde{c}) \leq q_s$ , with equality if  $\tilde{c}_s < 1$ , because then  $\check{c}_{ns} < 1$  eventually along the subsequence. So  $\tilde{c}$  solves Equation (10), and  $\tilde{c} = c$  by Lemma 3.1.2. Since  $[0, 1]^S$  is compact,  $\check{c}_n \rightarrow c$  almost surely. The second claim follows as in (a), with  $\check{c}_n$  in place of  $\hat{c}_n$ .

(c) Each offer rate is a ratio of sample frequencies of events defined by the realized cutoffs. Fix a school  $b$ , and let  $E_b$  be the set of types that rank  $b$  and rank no school above it with tie-breaker  $v(b)$ , so that  $\bar{c}_{nb}$  is the ratio of the averages over  $i \leq n$  of  $\mathbb{1}\{\theta_i \in E_b\} Z_i(b)$  and of  $\mathbb{1}\{\theta_i \in E_b\} A_i(b)$ . By Equation (4) in the market of size  $n$ , given  $\theta_i$ , each of  $Z_i(b)$  and  $A_i(b)$  indicates that  $R_i$  lies in a box in  $[0, 1]^V$  whose bounds are continuous functions of the realized cutoffs. By the multivariate Glivenko–Cantelli theorem, applied to the applicants of each type, the share of them whose lottery numbers lie in a box converges almost surely to its volume, uniformly over boxes. Since volumes are continuous in the bounds and  $\hat{c}_n \rightarrow c$  by Lemma A.4, the two averages converge almost surely to  $\mathbb{P}(\theta_i \in E_b, Z_i(b) = 1)$  and  $\mathbb{P}(\theta_i \in E_b, A_i(b) = 1)$  in the continuum. The second is positive by Assumption 2.3.2, because types that rank  $b$  first are in  $E_b$  and reach  $b$ . Since  $MID_{\theta b}^{v(b)} = 0$  for every  $\theta \in E_b$ , Lemma 3.2.1 shows that the ratio of the two is  $c_b$ , so  $\bar{c}_{nb} \rightarrow c_b$  almost surely.

For the offer probability, by Lemma A.4,  $\mathbb{P}_n(Z_i(s) = 1 \mid \theta_i = \theta) \rightarrow p_s(\theta; c)$  and  $\mathbb{P}_n(A_i(s) = 1 \mid \theta_i = \theta) \rightarrow \prod_V (1 - MID_{\theta s}^v)$ , which is positive because  $\theta$  reaches  $s$  with positive probability. Since  $Z_i(s) = 1$  implies  $A_i(s) = 1$ , the conditional probability is the ratio of the two, which converges to  $p_s(\theta; c) / \prod_V (1 - MID_{\theta s}^v)$ . By Equation (7), this limit is the right side of the display in (c).

(d) By assumption,  $MID_{\theta s}^{v(s)} = 0$  for every type  $\theta$  that ranks  $s$ , so the limit in (c) is  $c_s$ .

(e) As in the proof of Lemma A.4, with the event  $R_{iv(s)} \leq \hat{c}_{ns}$  in place of the offer,  $\mathbb{P}_n(R_{iv(s)} \leq \hat{c}_{ns} \mid \theta_i) \rightarrow \mathbb{P}(R_{iv(s)} \leq c_s) = c_s$  almost surely, by Assumption 2.1.1. The event depends only on types and lottery numbers, and lottery numbers are independent of types and potential outcomes, so conditioning also on  $(D_{0i}, D_{1i}, Y_{0i}, Y_{1i})$  leaves its probability, and so its limit  $c_s$ , unchanged.  $\square$

## B Additional simulation results

Each draw redraws the market and its lotteries. In both economies,  $\xi_i \sim N(0, 1)$ ; a random 85 percent of applicants enroll at the target school if and only if offered a seat there, and the rest never enroll there; and outcomes are  $Y_i = Y_{0i} + D_i$ .

In Example 2 of Abdulkadiroğlu et al. (2017), schools  $a$ ,  $b$ , and  $c$  each have  $m$  seats, and  $m$  applicants have each of the four types of Section 6. An applicant of type  $k \in \{1, 2, 3, 4\}$ , numbered

in the order listed there, has untreated outcome  $Y_{0i} = k - 1 + \xi_i$ .

In the random-preference economy, applicant  $i$  has ability  $a_i \sim N(0, 1)$  and utility  $u_{ij} = \delta_j + a_i \mathbb{1}\{j \in \{1, 4\}\} + e_{ij}$  for schools  $j = 1, \dots, 6$ , where  $\delta_j$  falls linearly from 0.5 to  $-0.5$  and each  $e_{ij}$  is standard Gumbel. The outside option has utility  $e_{i0} = 0.5$ , and  $i$  ranks the schools she prefers to it, in order of utility. School 4 is the target and school 1 the most popular. Untreated outcomes are  $Y_{0i} = a_i + \xi_i$ . Capacities are  $(150, 150, 250, 250, 300, 300)$  with 2,000 applicants and  $(20, 20, 35, 35, 45, 45)$  with 300. In a third tie-breaking design, schools 1–3 share one lottery, schools 4 and 5 share another, and school 6 has its own.

For the procedure of [Section 4.3](#), each school's waitlist order lists the applicants who applied to it by their lottery numbers there. The procedure examines all  $L = 15$  pairs of schools, groups common applicants using their lists, and joins pairs with  $w_{ss'} \geq \log(15/0.05)$ .

Each draw then estimates the effect at school 4 with types and lottery numbers, types alone, and ordered rejections, once with the true map and once with the recovered one. [Section 6](#) reports the results; [Tables B.1](#) and [3](#) use only STB and MTB.

Table B.1: Effect estimates by data regime in small markets

| Data                                                 | STB  |      |          | MTB  |      |          |
|------------------------------------------------------|------|------|----------|------|------|----------|
|                                                      | Mean | SD   | Coverage | Mean | SD   | Coverage |
| <i>AANP (2017) Example 2, 15 applicants per type</i> |      |      |          |      |      |          |
| Offer, no controls                                   | 2.98 | 0.47 | 0.01     | 2.68 | 0.45 | 0.04     |
| Waitlists                                            | 1.91 | 0.72 | 0.68     | 1.02 | 0.64 | 0.94     |
| Ordered rejections                                   | 1.01 | 0.79 | 0.82     | 1.02 | 0.64 | 0.94     |
| Types and lottery numbers                            | 1.00 | 0.69 | 0.83     | 1.01 | 0.46 | 0.90     |
| Types, solved cutoffs                                | 1.00 | 0.69 | 0.83     | 1.01 | 0.46 | 0.90     |
| Lottery numbers (qualification)                      | 1.06 | 2.12 | 0.93     | 1.12 | 1.26 | 0.96     |
| <i>Random preferences, 300 applicants</i>            |      |      |          |      |      |          |
| Offer, no controls                                   | 1.79 | 0.29 | 0.22     | 1.66 | 0.29 | 0.36     |
| Waitlists                                            | 1.19 | 0.34 | 0.93     | 1.02 | 0.33 | 0.94     |
| Ordered rejections                                   | 1.00 | 0.39 | 0.95     | 1.02 | 0.33 | 0.94     |
| Types and lottery numbers                            | 1.00 | 0.37 | 0.95     | 1.02 | 0.32 | 0.94     |
| Types, solved cutoffs                                | 1.00 | 0.37 | 0.95     | 1.02 | 0.32 | 0.94     |
| Lottery numbers (qualification)                      | 0.99 | 0.96 | 0.96     | 1.04 | 0.58 | 0.96     |

As in [Table 3](#), with 15 applicants of each type and 15 seats per school in Example 2 of [Abdulkadiroğlu et al. \(2017\)](#), and 300 applicants and 200 seats in the random-preference economy. In the 60-applicant market under STB, 46 of the 1,000 draws yield no estimate for ordered rejections or the designs that use types; statistics for those rows use the remaining 954 draws.